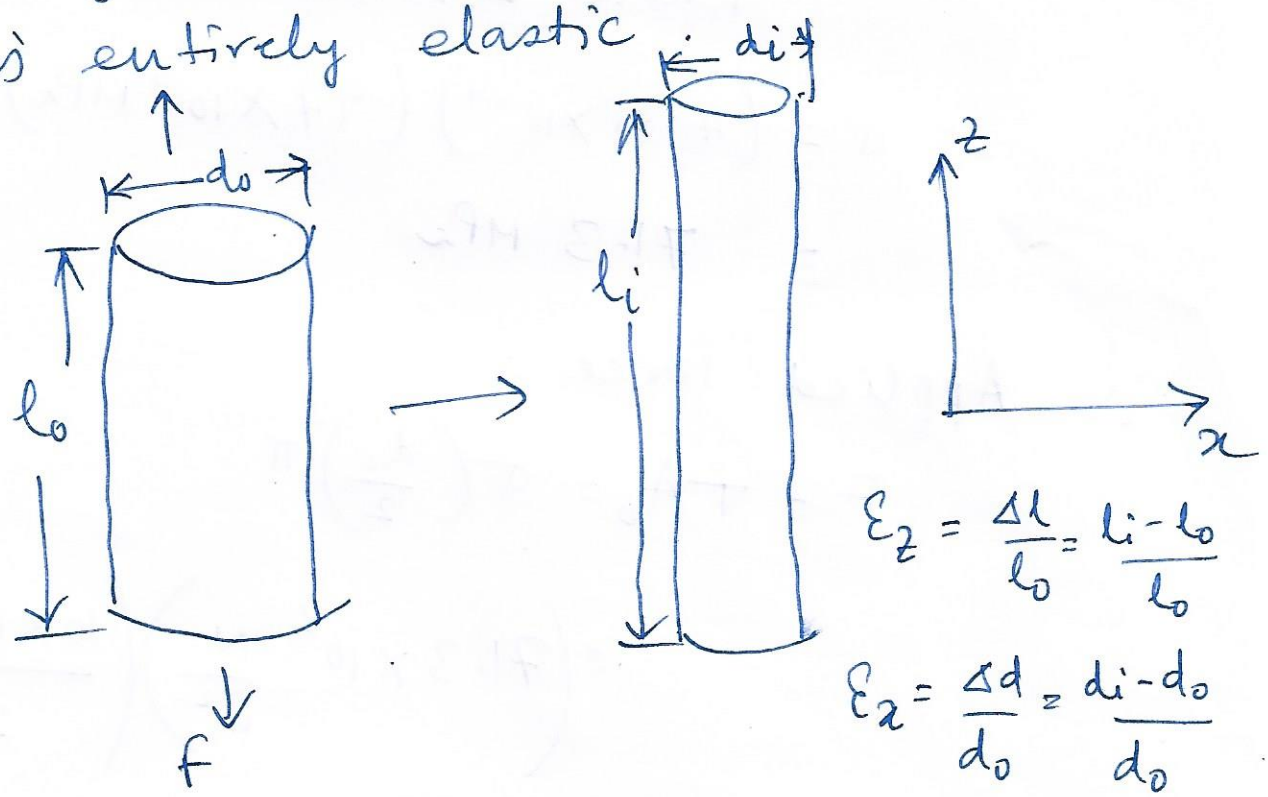
MO-201 Materials Science Tutorial

(1) Computation of load to produce specified diameter change

Q: A tensile stress is to be applied along the long axis of a cylindrical brass rod with dia. 10 mm. Determine the magnitude of load required to produce a 2.5×10^{-3} mm change in diameter if the deformation is entirely elastic

Ans:



$$\varepsilon_z = \frac{\Delta l}{l_o} = \frac{l_i - l_o}{l_o}$$

$$\varepsilon_z = \frac{\Delta d}{d_o} = \frac{d_i - d_o}{d_o}$$

Specimen will elongate along z-direction,
$\therefore$ reduction in diameter.

$\Delta d = 2.5 \times 10^{-3}$ mm in x-direction

$$\therefore \varepsilon_x = \frac{\Delta d}{d_0} = \frac{-2.5 \times 10^{-3}\ mm}{10\ mm} = -2.5 \times 10^{-4}$$

Value of Poisson's ratio for brass = 0.34

$$\therefore \varepsilon_z = -\frac{\varepsilon_x}{\nu} = -\frac{(-2.5 \times 10^{-4})}{0.34}$$

$$= 7.35 \times 10^{-4}$$

$\therefore$ Applied stress $\sigma = \varepsilon_z E$,

where $E = 97$ GPa (elastic modulus)

$$\therefore \sigma = (7.35 \times 10^{-4})(97 \times 10^3\ MPa)$$

$$= 71.3\ MPa$$

$\therefore$ Applied force

$$F = \sigma A_0 = \sigma \left(\frac{d_0}{2}\right)^2 \pi$$

$$= \left(71.3 \times 10^6\ \frac{N}{m^2}\right)\left(\frac{10 \times 10^{-3} m}{2}\right)^2 \pi$$

$$= \underline{\underline{5600\ N}}$$

(2) Ductility and True-Stress-at-Fracture

Q: A cylindrical specimen of steel with original dia. 12.8 mm is tensile tested to fracture & found to have an engineering fracture strength $\sigma_f = 460$ MPa. If its cross-sectional dia. at fracture is 10.7 mm, determine

(a) Ductility in terms of %RA

(b) True Stress at Fracture

Ans: (a) Ductility $\%RA = \left(\frac{A_o - A_f}{A_o}\right) \times 100$

$$= \frac{\pi\left(\frac{12.8}{2}\right)^2 - \pi\left(\frac{10.7}{2}\right)^2}{\pi\left(\frac{12.8}{2}\right)^2} \times 100$$

$$= 30\%$$

(b) $\sigma_T = \frac{F}{A_i}$, where, in this case, area is the fracture area A_f

First, calculate load at fracture from the fracture strength

$$F = \sigma_f A_o = \left(460 \times 10^6 \frac{N}{m^2}\right)\left(\pi \times \left(\frac{12.8}{2}\right)^2 mm^2\right)\left(\frac{1\ m^2}{10^6\ mm^2}\right)$$

$$= 59200\ N$$

$\therefore$ True stress at fracture

$$\sigma_T = \frac{F}{A_f} = \frac{59200\ N}{\pi\left(\frac{10.7}{2}\right)^2 mm^2 \left(\frac{1\ m^2}{10^6\ mm^2}\right)}$$

$$= 660\ MPa$$

(3) Materials Design Example: Specification for a Pressurized Cylindrical Tube

Q: Consider a thin-walled cylindrical tube having a radius 50 mm & wall thickness 2 mm that is to be used for transporting pressurized gas. Inside & Outside tube pressures are 20 atm & 0.5 atm (2.027 MPa & 0.057 MPa, respectively).

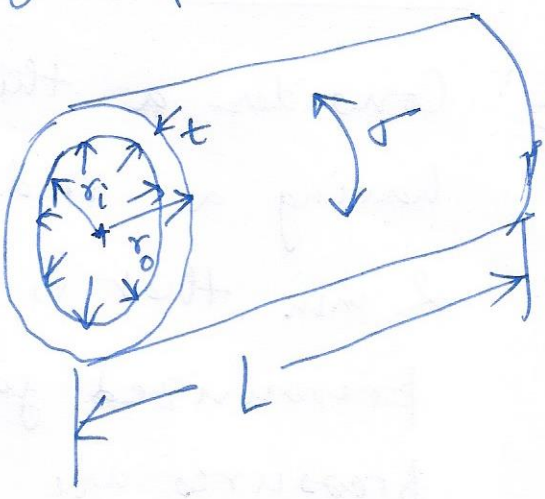
(a) Which of the metals/alloys listed in the table are suitable candidates?

Assume Safety factor = 4.0

(b) Also determine which alloy tube can be produced at lowest cost while satisfying the criterion in (a).

Alloy	*Yield Strength, σ_y (MPa)*	*Density, ρ (g/cm^3)*	*Unit Mass Cost, $\overline{c}$ ($US/kg)*
Steel	325	7.8	1.25
Aluminum	125	2.7	3.50
Copper	225	8.9	6.25
Brass	275	8.5	7.50
Magnesium	175	1.8	14.00
Titanium	700	4.5	40.00

Ans: We want to minimize possibility of tube getting plastically deformed.



Circumferential (hoop) stress (σ) given by

$$\sigma = \frac{r_i \Delta p}{t}$$

where Δp = pressure difference

t = thickness of tube wall

r_i = inner radius

To minimize plastic deformation possibility, we replace the circumferential stress by yield stress / factor of safety (N)

$$\therefore \frac{\sigma_y}{N} = \frac{r_i \Delta p}{t}$$

$$\Rightarrow \sigma_y = \frac{N r_i \Delta p}{t}$$

$$= \frac{(4.0)(50 \times 10^{-3}\ m)(2.027\ MPa - 0.057\ MPa)}{(2 \times 10^{-3}\ m)}$$

$$= \underline{197\ MPa}.$$

4 of the 6 alloys in the table have YS > 197 MPa, i.e.,

Steel, Copper, Brass, Ti

(b) To find the cost of the tube for each alloy, calculate volume $V = AL$

$$= \pi(r_o^2 - r_i^2)L$$

Since $r_o = r_i + t$

$$\therefore V = \pi\left[(r_i + t)^2 - r_i^2\right]L$$

$$= \pi\left(r_i^2 + 2r_i t + t^2 - r_i^2\right)L$$

$$= \pi\left(2r_i t + t^2\right)L$$

Let us assume a tube of length $L = 1.0\ m$

$$\therefore V = \pi\left[(2)(50\times10^{-3}\ m)(2\times10^{-3}\ m) + (2\times10^{-3}\ m)^2\right](1\ m)$$

$$= 6.28\times10^{-4}\ m^3$$

$$= 628\ cm^3$$

Now, calculate mass of each alloy = vol. x density & then multiply by unit mass cost.

e.g.:

$$\text{Cost (Steel)} = \left[\frac{(628\ \text{cm}^3)(7.8\ \text{g/cm}^3)}{(1000\ \text{g/kg})}\right] \times (1.25\ \$/\text{kg})$$

$$= \$6.10$$

Similarly, for other 3 alloys →

Alloy	Cost ($)
Steel	6.10
Cu	35.00
Brass	40.00
Ti	113.00

∴ Steel is the least expensive alloy for the pressurized tube.